ri-district-vs-charter-performance-analysis

RM

2026-02-08

```r
setwd("/Users/ryanmunroe/Downloads")
```

Library / Package Setup

```r
library(tidyr)
library(tidyverse)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.0     ✔ purrr     1.0.2
## ✔ forcats   1.0.0     ✔ readr     2.1.5
## ✔ ggplot2   4.0.0     ✔ stringr   1.6.0
## ✔ lubridate 1.9.4     ✔ tibble    3.2.1
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(broom)
library(ggplot2)
```

Load Data

```r
school_indicator_data <- read.csv("school_indicator_data.csv")

lea_indicator_data <- read.csv("lea_indicator_data.csv")

csi_schools <- read.csv("csi_schools_data.csv")

atsi_groups <- read.csv("atsi_groups_data.csv")

tsi_groups <- read.csv("tsi_groups_data.csv")

lea_low_performing_subgroups <- read.csv("lea_low_performing_subgroups.csv")
```

Fixing Column Labels (run only once)

```r
names(school_indicator_data) <- school_indicator_data[1, ]
school_indicator_data <- school_indicator_data[-1, , drop = FALSE]
rownames(school_indicator_data) <- NULL

names(lea_indicator_data) <- lea_indicator_data[1, ]
lea_indicator_data <- lea_indicator_data[-1, , drop = FALSE]
rownames(lea_indicator_data) <- NULL

names(csi_schools) <- csi_schools[1, ]
csi_schools <- csi_schools[-1, , drop = FALSE]
rownames(csi_schools) <- NULL

names(atsi_groups) <- atsi_groups[1, ]
atsi_groups <- atsi_groups[-1, , drop = FALSE]
rownames(atsi_groups) <- NULL

names(tsi_groups) <- tsi_groups[1, ]
tsi_groups <- tsi_groups[-1, , drop = FALSE]
rownames(tsi_groups) <- NULL

names(lea_low_performing_subgroups) <- lea_low_performing_subgroups[1, ]
lea_low_performing_subgroups <- lea_low_performing_subgroups[-1, , drop = FALSE]
rownames(lea_low_performing_subgroups) <- NULL
```

```r
# unique(school_indicator_data$District)
# unique(school_indicator_data$School)
```

```r
school_indicator_data$`School Code` <- as.character(school_indicator_data$`School Code`)

selected_charter_schools <- c("41190", "41607", "28614", "41605", "41606", "28609", "41604", "28615", "07135", "3
9601", "26602", "08603", "08605", "08601", "08604", "08602", "08606", "84601", "39603", "48601", "48602", "2660
1", "32601", "69601", "51601", "51602", "51603", "83601", "28607", "39602", "79601", "04602", "46602", "28611", "
23601", "97601", "28613", "04601", "28606", "28608")

charter_df <- school_indicator_data[
  school_indicator_data$`School Code` %in% selected_charter_schools,
]

charter_df$charter_school_name <- charter_df$`School Name`

selected_lea_public_schools <- c("16108","17112","30104","30105","35127","10123","31107","32108","01109","3611
1","07120","36106","03105","24106","07127","23114","23113","28162","16103","06116","33108","33110","35138","2815
1","28130","04106","31109","38111","33105","35137","35133","35142","06121","97101","28127","28194","13104","3211
2","38106","36104","36113","36103","06109","07129","23105","18101","28189","28153","26105","03107","07116","3513
9","16109","07121","39123","39131","39132","35119","01105","35135","39101","28161","26119","35128","30102","2814
9","07137","98109","06112","27104","07115","96113","26109","09102","24107","10106","15101","15102","38107","3810
4","08110","16112","26103","08115","10113","39128","96112","39120","28103","28157","17109","17117","17110","3512
3","17106","26107","26121","38105","28160","32113","09107","27111","97103","19111","97102","28150","96107","1012
5","20102","20104","20103","28143","28145","26111","01104","16111","23103","08116","23108","24105","30106","2511
0","25108","25109","17113","35101","07103","35104","31104","07136","10124","35132","07123","32107","35134","3110
6","28165","33107","99102","99101","27106","27112","39133","26110","01103","28193","04112","04118","31108","2814
2","98105","10122","28138","28156","96106","28147","21111","26106","10116","26122","06104","28121","10114","2818
1","19106","09103","07125","28115","98106","08112","03108","28199","08107","01106","01108","39118","24112","0611
7","22101","28119","32114","16106","03101","03109","03104","04104","12101","35131","04108","28139","24109","9810
8","98101","98103","26118","98104","21107","28164","30103","96105","08109","35136","06119","07104","07113","0712
6","08114","08121","26116","23112","04115","24111","28113","24108","28163","36109","28187","35121","16114","0910
6","10112","07118","07130","10109","10104","07117","26120","04101","10117","28137","97105","97106","26113","2310
9","26115","13103","19107","23110","26125","28134","21106","09105","07122","08108","19117","19114","09108","0712
4","28135","07119","07128","39119","28182","16110","38109","35114","24110","23111")

lea_public_df <- school_indicator_data [
  school_indicator_data$`School Code` %in% selected_lea_public_schools,
  ]

lea_public_df$public_school_name <- lea_public_df$`School Name`
```

```r
na_counts <- charter_df %>%
  summarise(
    STAR_score_NA = sum(is.na(`2018 Star Rating`)),
  )

print(na_counts)
```

```
##   STAR_score_NA
## 1             0
```

STAR Cleaning

```r
star_columns <- names(charter_df) %>%
  stringr::str_subset("Star Rating")

star_columns
```

```
## [1] "2018 Star Rating" "2019 Star Rating" "2022 Star Rating" "2023 Star Rating"
## [5] "2024 Star Rating" "2025 Star Rating"
```

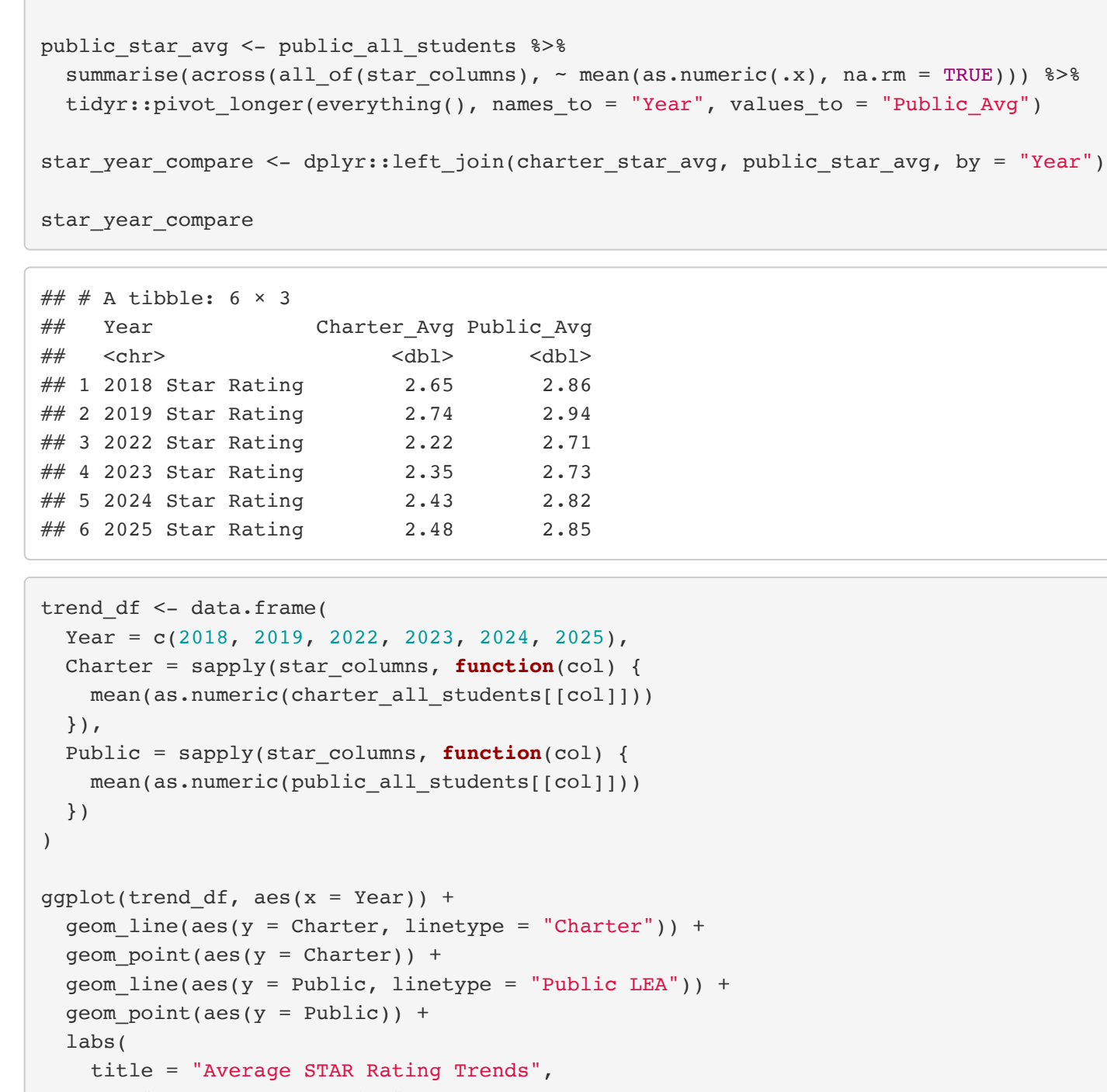
```r
charter_df[star_columns] <- lapply(charter_df[star_columns], as.numeric)
lea_public_df[star_columns] <- lapply(lea_public_df[star_columns], as.numeric)

charter_all_students <- charter_df %>%
  dplyr::filter(Group == "All Students")

public_all_students <- lea_public_df %>%
  dplyr::filter(Group == "All Students")

charter_all_students <- charter_all_students %>%
  dplyr::filter(dplyr::if_all(dplyr::all_of(star_columns), ~ !is.na(.)))

public_all_students <- public_all_students %>%
  dplyr::filter(dplyr::if_all(dplyr::all_of(star_columns), ~ !is.na(.)))
```

Average Star Comparison by Year

```r
charter_star_avg <- charter_all_students %>%
  summarise(across(all_of(star_columns), ~ mean(as.numeric(.x), na.rm = TRUE))) %>%
  tidyr::pivot_longer(everything(), names_to = "Year", values_to = "Charter_Avg")

public_star_avg <- public_all_students %>%
  summarise(across(all_of(star_columns), ~ mean(as.numeric(.x), na.rm = TRUE))) %>%
  tidyr::pivot_longer(everything(), names_to = "Year", values_to = "Public_Avg")

star_year_compare <- dplyr::left_join(charter_star_avg, public_star_avg, by = "Year")

star_year_compare
```

```
## # A tibble: 6 × 3
##   Year             Charter_Avg Public_Avg
##   <chr>                  <dbl>      <dbl>
## 1 2018 Star Rating        2.65       2.86
## 2 2019 Star Rating        2.74       2.94
## 3 2022 Star Rating        2.22       2.71
## 4 2023 Star Rating        2.35       2.73
## 5 2024 Star Rating        2.43       2.82
## 6 2025 Star Rating        2.48       2.85
```

```r
trend_df <- data.frame(
  Year = c(2018, 2019, 2022, 2023, 2024, 2025),
  Charter = sapply(star_columns, function(col) {
    mean(as.numeric(charter_all_students[[col]]))
  }),
  Public = sapply(star_columns, function(col) {
    mean(as.numeric(public_all_students[[col]]))
  })
)

ggplot(trend_df, aes(x = Year)) +
  geom_line(aes(y = Charter, linetype = "Charter")) +
  geom_point(aes(y = Charter)) +
  geom_line(aes(y = Public, linetype = "Public LEA")) +
  geom_point(aes(y = Public)) +
  labs(
    title = "Average STAR Rating Trends",
    y = "Average STAR Rating"
  ) +
  theme_minimal()
```



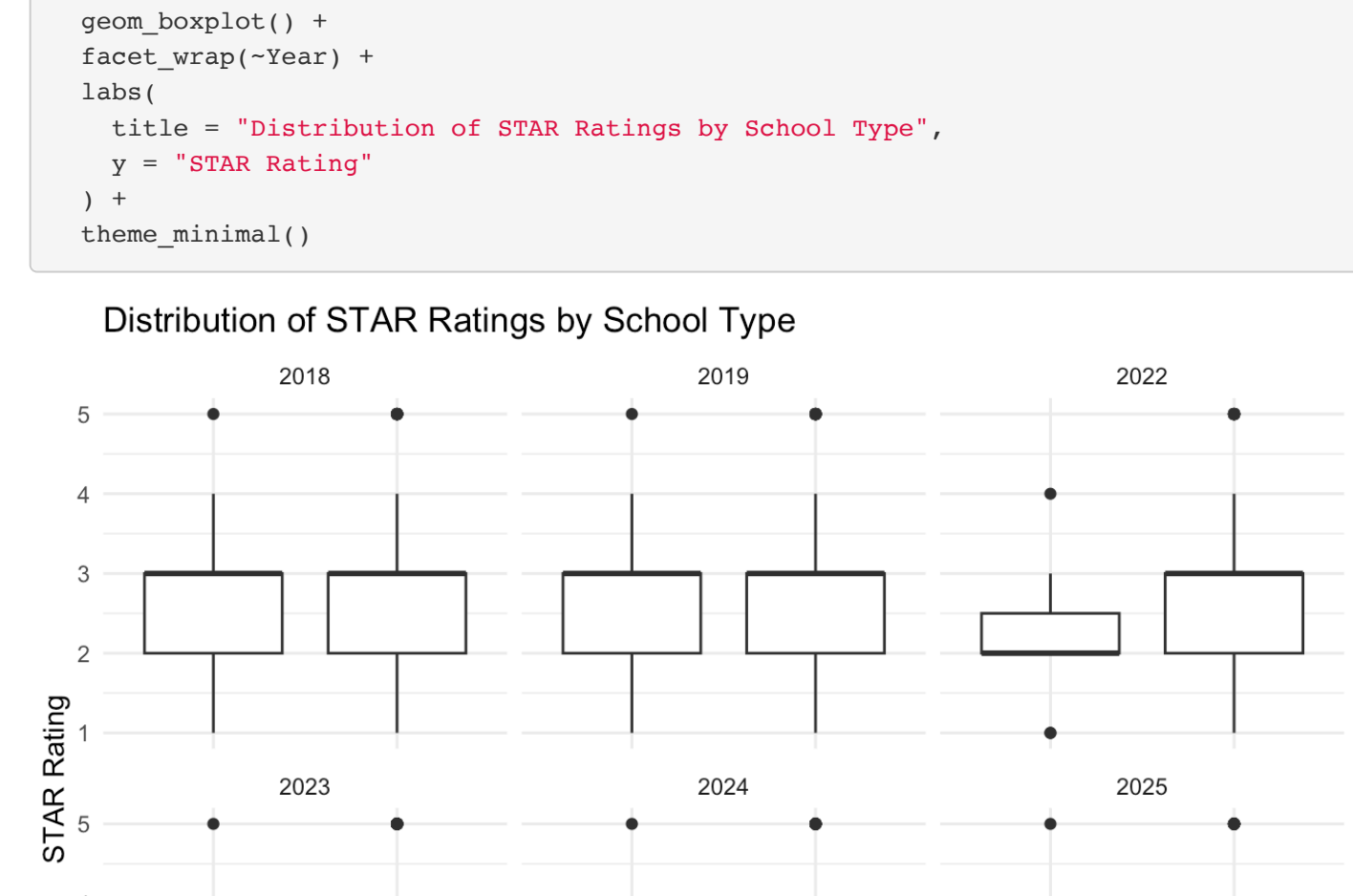
Testing by Year (if there is a significant difference in testing by year)

```r
for (col in star_columns) {

  year <- stringr::str_extract(col, "\\d{4}")

  charter_values <- as.numeric(charter_all_students[[col]])
  public_values  <- as.numeric(public_all_students[[col]])

  test <- t.test(charter_values, public_values)

  cat("\n============================\n")
  cat("STAR Rating Year:", year, "\n")
  cat("============================\n")

  print(test)
}
```

```
## 
## ============================
## STAR Rating Year: 2018 
## ============================
## 
##  Welch Two Sample t-test
## 
## data:  charter_values and public_values
## t = -0.96143, df = 26.757, p-value = 0.3449
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.6483296  0.2347288
## sample estimates:
## mean of x mean of y 
##  2.652174  2.858974 
## 
## 
## ============================
## STAR Rating Year: 2019 
## ============================
## 
##  Welch Two Sample t-test
## 
## data:  charter_values and public_values
## t = -0.89194, df = 26.482, p-value = 0.3805
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.6498264  0.2562924
## sample estimates:
## mean of x mean of y 
##  2.739130  2.935897 
## 
## 
## ============================
## STAR Rating Year: 2022 
## ============================
## 
##  Welch Two Sample t-test
## 
## data:  charter_values and public_values
## t = -3.1528, df = 32.677, p-value = 0.003456
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.8025909 -0.1728829
## sample estimates:
## mean of x mean of y 
##  2.217391  2.705128 
## 
## 
## ============================
## STAR Rating Year: 2023 
## ============================
## 
##  Welch Two Sample t-test
## 
## data:  charter_values and public_values
## t = -2.3021, df = 32.621, p-value = 0.02786
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.71347267 -0.04386661
## sample estimates:
## mean of x mean of y 
##  2.347826  2.726496 
## 
## 
## ============================
## STAR Rating Year: 2024 
## ============================
## 
##  Welch Two Sample t-test
## 
## data:  charter_values and public_values
## t = -2.0203, df = 29.155, p-value = 0.05262
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.767524447  0.004611032
## sample estimates:
## mean of x mean of y 
##  2.434783  2.816239 
## 
## 
## ============================
## STAR Rating Year: 2025 
## ============================
## 
##  Welch Two Sample t-test
## 
## data:  charter_values and public_values
## t = -1.804, df = 27.373, p-value = 0.08224
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.80431454  0.05143457
## sample estimates:
## mean of x mean of y 
##  2.478261  2.854701
```

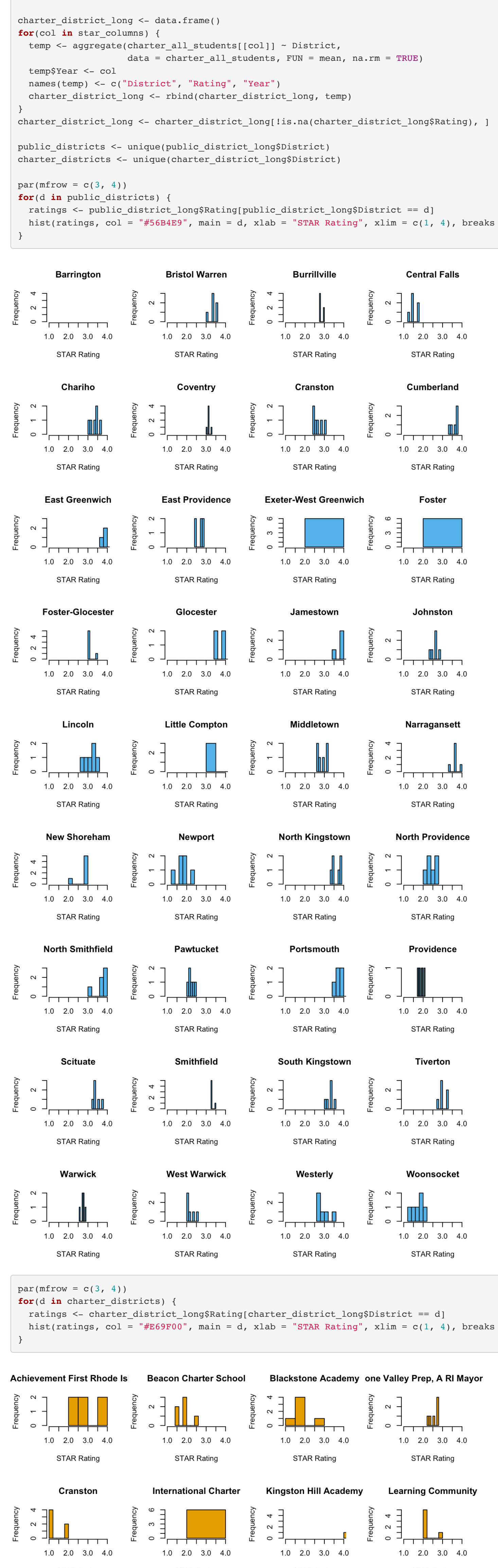
(LOW SAMPLE SIZE FOR CHARTER SCHOOLS / LIMIT TO LOW SCORE OUTLIERS)

```r
charter_all_students$School_Type <- "Charter"
public_all_students$School_Type  <- "Public LEA"

combined_df <- bind_rows(charter_all_students, public_all_students)

combined_long <- combined_df %>%
  pivot_longer(
    cols = all_of(star_columns),
    names_to = "Year",
    values_to = "Star"
  ) %>%
  mutate(
    Year = gsub(" Star Rating", "", Year),
    Star = as.numeric(Star)
  )

ggplot(combined_long, aes(x = School_Type, y = Star)) +
  geom_boxplot() +
  facet_wrap(~Year) +
  labs(
    title = "Distribution of STAR Ratings by School Type",
    y = "STAR Rating"
  ) +
  theme_minimal()
```

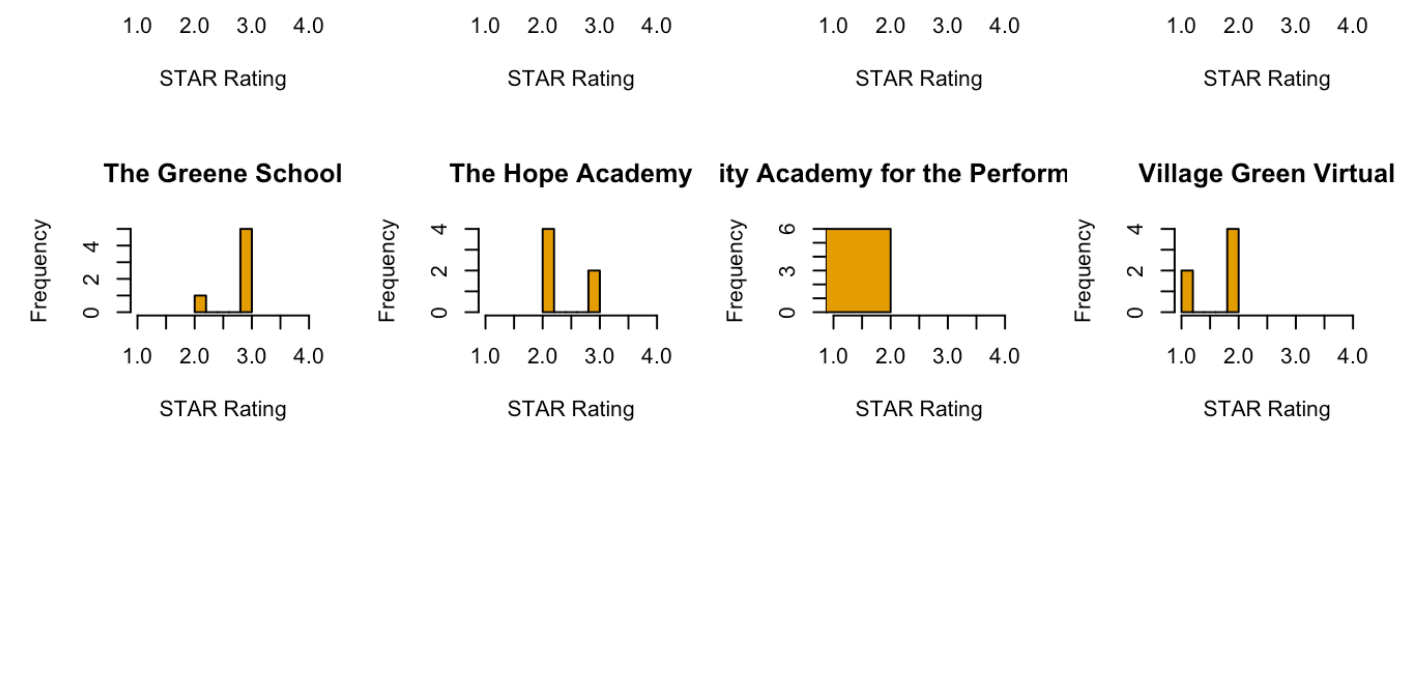


Comparing by district

```r
public_district_long <- data.frame()
for(col in star_columns) {
  temp <- aggregate(public_all_students[[col]] ~ District, data = public_all_students,
                    FUN = mean, na.rm = TRUE)
  temp$Year <- col
  names(temp) <- c("District", "Rating", "Year")
  public_district_long <- rbind(public_district_long, temp)
}
public_district_long <- public_district_long[!is.na(public_district_long$Rating), ]

charter_district_long <- data.frame()
for(col in star_columns) {
  temp <- aggregate(charter_all_students[[col]] ~ District,
                    data = charter_all_students, FUN = mean, na.rm = TRUE)
  temp$Year <- col
  names(temp) <- c("District", "Rating", "Year")
  charter_district_long <- rbind(charter_district_long, temp)
}
charter_district_long <- charter_district_long[!is.na(charter_district_long$Rating), ]

public_districts <- unique(public_district_long$District)
charter_districts <- unique(charter_district_long$District)

par(mfrow = c(3, 4))
for(d in public_districts) {
  ratings <- public_district_long$Rating[public_district_long$District == d]
  hist(ratings, col = "#56B4E9", main = d, xlab = "STAR Rating", xlim = c(1, 4), breaks = 5)
}
```

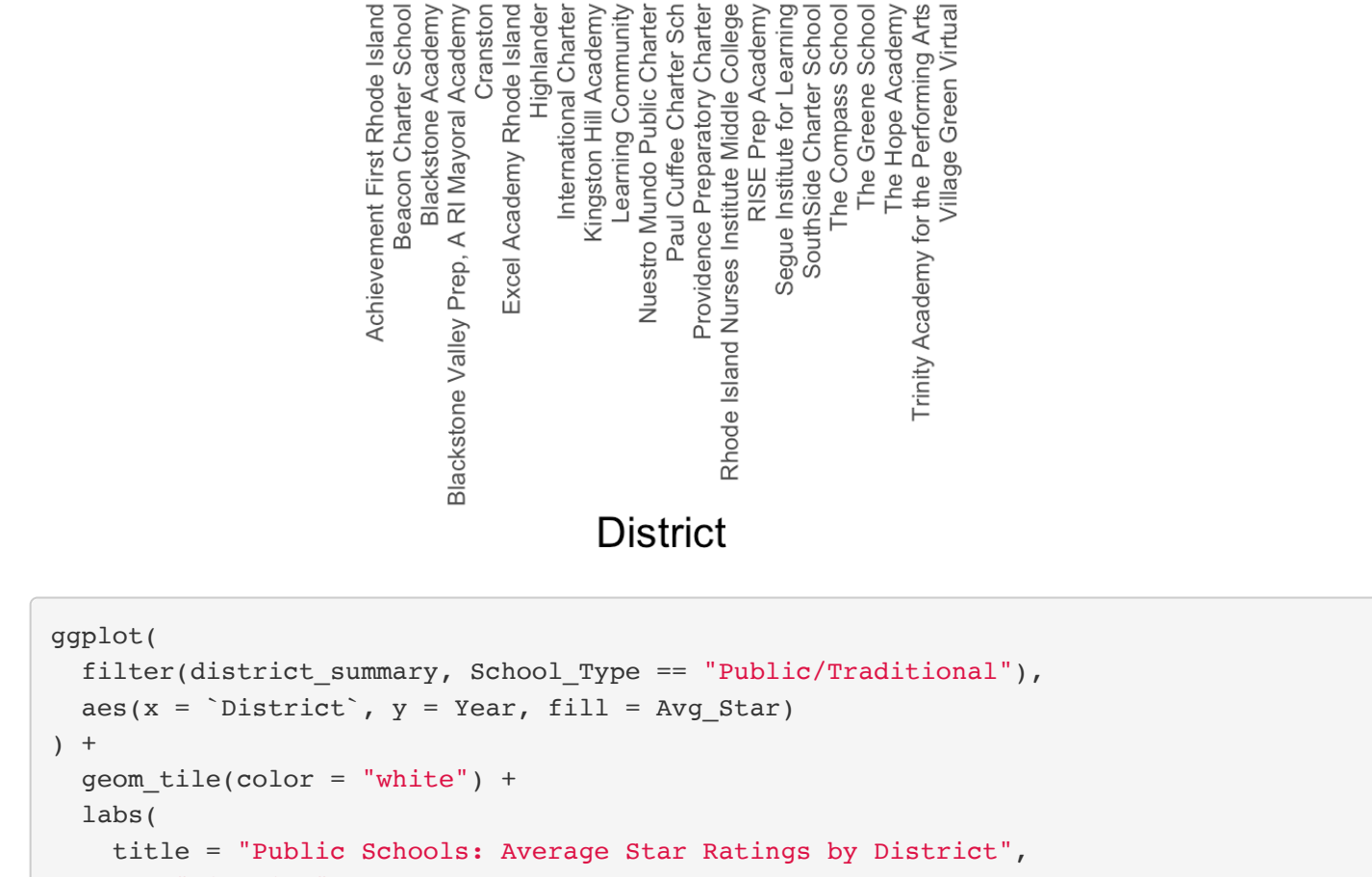


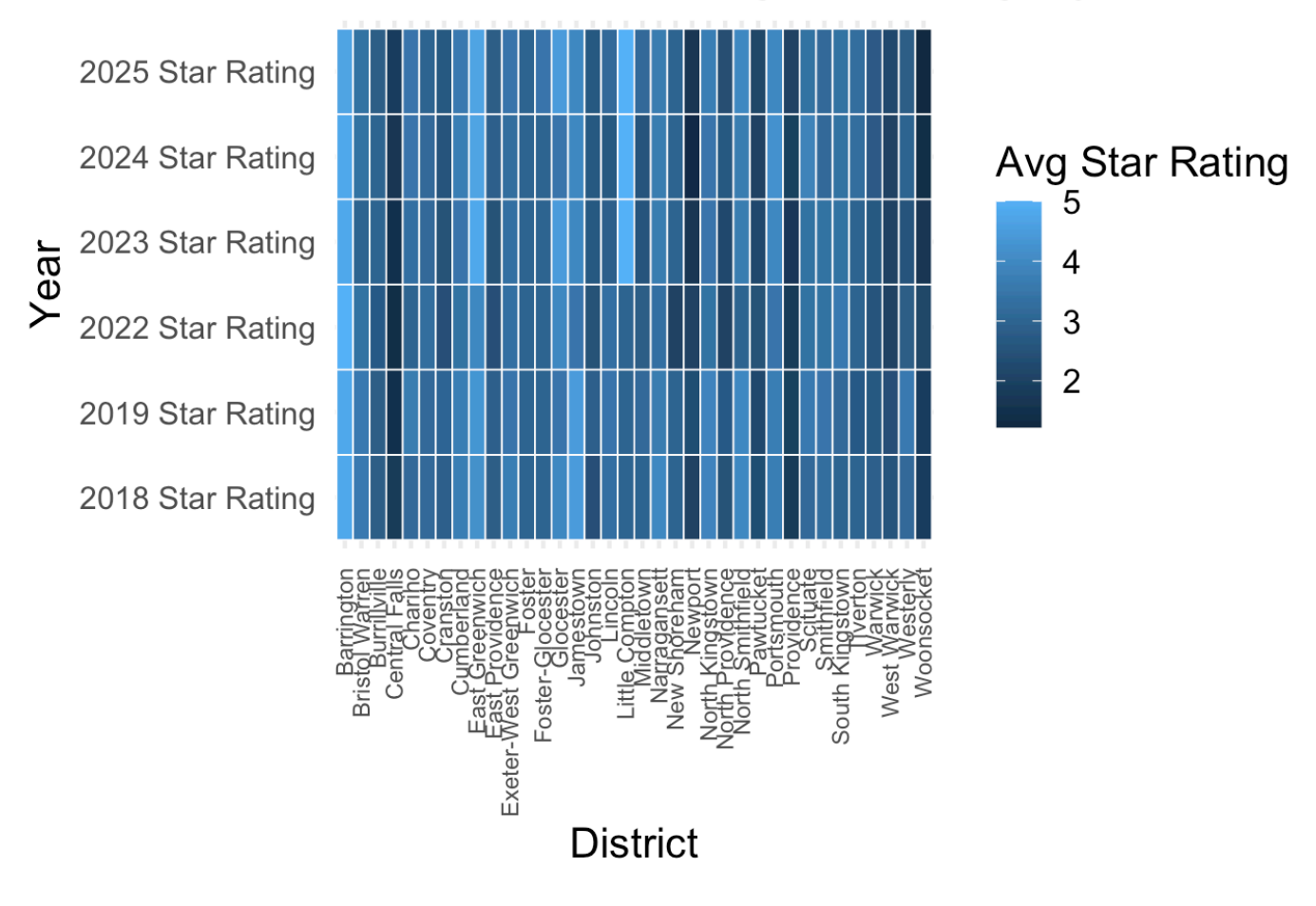
```r
par(mfrow = c(3, 4))
for(d in charter_districts) {
  ratings <- charter_district_long$Rating[charter_district_long$District == d]
  hist(ratings, col = "#E69F00", main = d, xlab = "STAR Rating", xlim = c(1, 4), breaks = 5)
}
```



Heatmap of STAR scores by district for public and charter schools

```r
charter_df <- charter_df %>%
  mutate(School_Type = "Charter")

lea_public_df <- lea_public_df %>%
  mutate(School_Type = "Public/Traditional")

heatmap_df <- bind_rows(charter_df, lea_public_df) %>%
  filter(Group == "All Students")

heatmap_df <- heatmap_df %>%
  mutate(across(contains("Star Rating"), as.numeric))

heatmap_long <- heatmap_df %>%
  pivot_longer(
    cols = contains("Star Rating"),
    names_to = "Year",
    values_to = "StarScore"
  )

district_summary <- heatmap_long %>%
  group_by(`District`, School_Type, Year) %>%
  summarize(
    Avg_Star = mean(StarScore, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(
  filter(district_summary, School_Type == "Charter"),
  aes(x = `District`, y = Year, fill = Avg_Star)
) +
  geom_tile(color = "white") +
  labs(
    title = "Charter Schools: Average Star Ratings by District",
    x = "District",
    y = "Year",
    fill = "Avg Star Rating"
  ) +
  theme_minimal(base_size = 16) +
  theme(
    axis.text.y = element_text(size = 12),
    axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, size = 9),
    plot.title  = element_text(size = 15, face = "bold", hjust = 1),
  )
```



```r
ggplot(
  filter(district_summary, School_Type == "Public/Traditional"),
  aes(x = `District`, y = Year, fill = Avg_Star)
) +
  geom_tile(color = "white") +
  labs(
    title = "Public Schools: Average Star Ratings by District",
    x = "District",
    y = "Year",
    fill = "Avg Star Rating"
  ) +
  theme_minimal(base_size = 16) +
  theme(
    axis.text.y = element_text(size = 12),
    axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, size = 9),
    plot.title  = element_text(size = 15, face = "bold")
  )
```